

Course milestone M7: First Reproducible Analysis (+ Conference Application)

HONR 46400 · Evidence-Driven Research

Start here

Your main file is [lastname_m07.pdf](#). Build it in this order:

1. **Your carry-forward record, at the very top.** Open my review email and your research action plan side by side. Answer every numbered request for this milestone (each has an ID like [IR-M07-01](#)) and every step your plan sets for this milestone. Next to each one, write **Done** and exactly where I can find it, or **Not adopted**, with your reason and what you chose instead. Several items can point to the same piece of work.
2. **Then your milestone work and files.** The book milestone below explains the research work. The complete course checklist, the files to hand in and the rubric are in this milestone's assignment on Brightspace: check them before you submit.
3. **The Expo work** in *What this course adds*, below.

How this counts. Your answer to the action plan is worth 10 points of the rubric, scored for you individually. Each numbered request left unanswered costs 5 points, up to 20. Only requests and plan steps I sent you at least 48 hours before the deadline count; anything later moves to the next milestone. A step that is also a numbered request is counted once, as a request. If I sent you no numbered requests, or no plan steps apply, say so in one line. Not sure what applies to you? Write to me before the deadline; that is exactly what I am here for.

Submit it on Brightspace, under **Assignments** then **M07**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 7: Your first reproducible analysis** (checkpoint, version 1) of the course book, EDR|AI.

What this course adds

The conference submission

Three things leave your hands this week on the conference's calendar rather than on your research schedule, and the book milestone has none of them.

1. **Your final title and abstract.** Start from the title and abstract you drafted at Milestone 6, the comments on them in your review, and what you and your mentor decided about them at your first mentor meeting. Revise both into what you submit. Where the draft said what your analysis will estimate, the final version reports what it did estimate.

Follow the conference's limits: the portal caps the title at 125 characters and the abstract at 2,000 characters, spaces included in both. Capitalize the first letter of each major word in the title, and put no author name inside the abstract.

Your title highlights your research problem and your answer. A reader who sees only the title should know what you studied and what you found, bounded exactly the way the abstract bounds it, and an expected answer stays an expectation in the title too.

Your abstract is one structured paragraph. Write it as a single paragraph, with no line breaks, that moves through four parts in this order. The order of the parts carries the structure.

1. **Question, answer, context, contribution: four sentences.** The first sentence is your research question. The second is your answer to it: your verified result, worded no more strongly than your route licenses; if your result is not final yet, the answer you expect, worded plainly as an expectation (see part 4). The third gives the background context, including the

main conclusion of the literature on your question, then the research gap that conclusion leaves, your primary objective, and your main hypothesis: what you expected to find before your analysis ran. State the literature's conclusion from sources you have read and verified, never from an AI summary. The fourth states your contribution to the field: how your answer adds value to that main conclusion of the literature, for example by extending it, qualifying it, testing it in a new setting, or challenging it.

2. **Your data.** A description of your dataset: its source, units, coverage, and the measures you built. Where you can, add one or two results from your exploratory data analysis (EDA) with the insight each one gives.

In the book: Lesson 20, Data Provenance and Data Quality, shows how to say where each value comes from and what it is a value of; its worked example traces a county turnout figure back to the clerk's certified canvass and finds three different turnout rates behind three denominators. Lesson 21, Measurement and Operationalization, shows how to describe the measures you built, from concept to construct to indicator. An EDA insight is exploratory, a pattern you noticed after seeing the data, so label it that way and never present it as a test (Lesson 12, Preregistration and the pre-analysis plan).

Example, built from Lesson 21's worked example: "The data are 960 survey answers from employees at 16 work sites, 60 per site, eight that piloted flexible scheduling and eight that kept fixed shifts. Engagement is measured through one facet, willingness to recommend the workplace, as the mean answer to a single 1–7 agreement item; re-asking a random tenth of respondents two weeks later returned closely matching answers."

3. **Your study design.** The design, the sample size, the tools that collected the data, and the analytical technique your pipeline executes.

In the book: Lesson 10, Model, Inquiry, Data Strategy, and Answer Strategy, gives you the words for this part. Your data strategy is who was sampled, who got which condition, and how each outcome was measured: your sample and your data-collection tools. Your answer strategy is the estimate you computed and how you stated its uncertainty: your analytical technique. Lesson 12, Declaring and Diagnosing a Research Design, shows why a sample size is justified, not just reported: in its worked example, 12 shoppers per version detect a real 8-second gain in under 10% of simulated runs, and 400 per version in over 90%. Lesson 11, Dependence, and what it costs, shows why the sample size you report counts independent units: 60 flats in 12 buildings, treated as 60 independent flats, give an interval that catches the truth far less often than it promises. The pathway lessons of Studio 5 name the design your route declares.

Example, built from Lesson 12's worked example: "In a randomized experiment, 800 online shoppers were assigned by coin flip to check out with or without a one-tap reorder button, 400 per version, a size a simulation diagnosis showed detects an 8-second gain in over 90% of runs. Seconds from cart to payment were logged for every completed checkout, and the effect was estimated as the difference in group means, reported with its 95% interval."

4. **Your findings and what they mean.** Your primary finding, stated succinctly and objectively, with its numbers: the data points, estimate, or effect size, and the uncertainty around it. No promise to discuss it later. You can present expected results: the Expo accepts ongoing research with preliminary results or anticipated findings, so if your result is not final yet, present the result you expect instead. Label it as expected ("I expect", never "I find") and say what the expectation rests on: your hypothesis, the literature, your EDA, or your first run. A number you have not computed may appear only as an expectation with its source, never as a result. When the result is verified, update the same submission before the Expo's final-updates deadline. Then the broader impact and usefulness of the result, its main limitations, and its future direction.

One rule is non-negotiable: the abstract may use only the claims your current verified result licenses. Expected results follow the same rule: label each one as expected, and expect nothing your route could not license. Your result is verified this week and not yet audited, so your wording has to survive next week's robustness work without becoming false. The words are yours: the Expo's AI guidance does not allow AI to generate your abstract or any full section of your presentation, and asks you to disclose any generative AI use on your work.

Submit the title and the abstract exactly as you entered them in the portal, each with its character count, followed by a short list of what changed from your draft and which review comment, numbered request, or meeting decision each change answers.

2. **The application that puts your name on the programme.** Apply through OURConnect, Purdue's undergraduate research portal, following the conference's own Guidelines, Apply page, and abstract-submission how-to guide.

- Log in at OURConnect with BoilerKey, then open **Conference** and the **New Abstract** entry for the Fall Expo.

- Paste your title and abstract unchanged, allow the abstract to be published in the booklet, choose the **in-person** format and the **Poster** presentation type, and select at least the two poster-session slots the form requires, and every slot you can genuinely make. Do not ask to be considered for a research talk: a talk replaces the poster this course grades.
- Choose the primary research category that describes your project and the academic review unit the Expo's FAQ points you to; ask your mentor if it is not obvious. Optional program and funding fields stay blank unless one genuinely applies.
- List yourself as the presenting author, and your faculty mentor of record, Davi Cordeiro Moreira, dcordeir@purdue.edu, as the primary mentor. Click **Confirm** after each email. An application without a mentor does not stand.
- Click **Submit**, not **Save without submitting**: a saved abstract stays Pending and has not been sent. Then check that the automated confirmation email arrived and that **My Abstracts** shows it as Submitted. If either is missing, resubmit from an incognito browser window, as the Expo instructs.

Your mentor approves the submission in the portal. If a change is needed, update the same submission rather than starting a new one.

3. Your proof of application. Save the confirmation email as a PDF, add a screenshot of your row on My Abstracts showing your title, Presentation ID, and status, and submit them. Without that confirmation there is no poster session for you to present at, however good the rest of your project is. The deadline belongs to the conference, not to this course, and it is posted on your course platform.

Your Round 01 mentor meeting, confirmed

The meeting you requested at Milestone 4 has to have happened by now, inside the Round 01 window your course platform posts. This milestone collects the proof that it did.

Write one short section into your milestone document naming the day and hour of the meeting, who attended, and the two or three decisions that came out of it, including what was decided about your title and abstract. For each decision, say what you changed. Where you kept your course, say that plainly and say why the meeting did not move you. A meeting that changed nothing and cannot say why was a status report.

Record it the day it happens rather than reconstructing it later, and remember that the conversation itself is human work: your AI Research Ledger gets a row only for what you delegated afterwards.

Milestone 7: Your first reproducible analysis

Open the milestone workbook in Colab

This chapter closes Studio 7: Produce a reproducible first analysis. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A route-specific result with an uncertainty statement, a restart-and-run-all verification, an environment record, and a claim-to-output check.

What you bring

You also carry forward the last milestone's artifact: **Milestone 6: Your data and measurement, governed**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 22: AI as Programmer: your first computed number, its delegated code read line by line, and the sentence naming what the number does not cover.
- Lesson 23: AI as Analytical Assistant: your headline estimate with its uncertainty statement and cycle log, two independently re-derived numbers, a placebo run where licensed, the clean-restart record, the environment record, and the claim-to-output trace.

The practice

1. Write the analysis you declared, and only that analysis, before looking at anything else.
2. Produce the result with its uncertainty statement, in the form your Contract specified.

3. Restart and run everything from a clean state; if the numbers move, the pipeline is the finding.
4. Record your environment, and check that every claim you plan to make traces to a specific output.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** What you send to a tool during analysis is a disclosure decision each time.
- **Evidence, provenance, and reproducibility.** Every number in your result traces to a data cell and a line of code.
- **AI activity, verification, and human decisions.** Delegate the writing of code freely; verify every returned number against the data yourself.
- **Uncertainty, claim boundary, and revision history.** A result reported without uncertainty is not yet a result.

Where this milestone sits

You arrived carrying Milestone 6: Your data and measurement, governed. Milestone 8 exists to attack this result; the cleaner your record here, the more the attack can teach you. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Failing to reproduce your own analysis sends you back here before any stress-testing is meaningful.

The book's own checks

Your course rubric on Brightspace scores these checks inside its rows, so use this table to test your work before you hand it in. The points below are the book's, not your course score.

Each row scores **0, 1, or 2. 10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: A route-specific result with an uncertainty statement, a restart-and-run-all verification, an	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

#	Criterion	0	1	2
	environment record, and a claim-to-output check			

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` → `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` → `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.